Demo_3_PopulationStratification_2025

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```
knitr::opts_chunk$set(echo = FALSE)
knitr::opts_chunk$set(message = FALSE)
knitr::opts_chunk$set(warning = FALSE)

# Load relevant libraries
require(ggplot2)
```

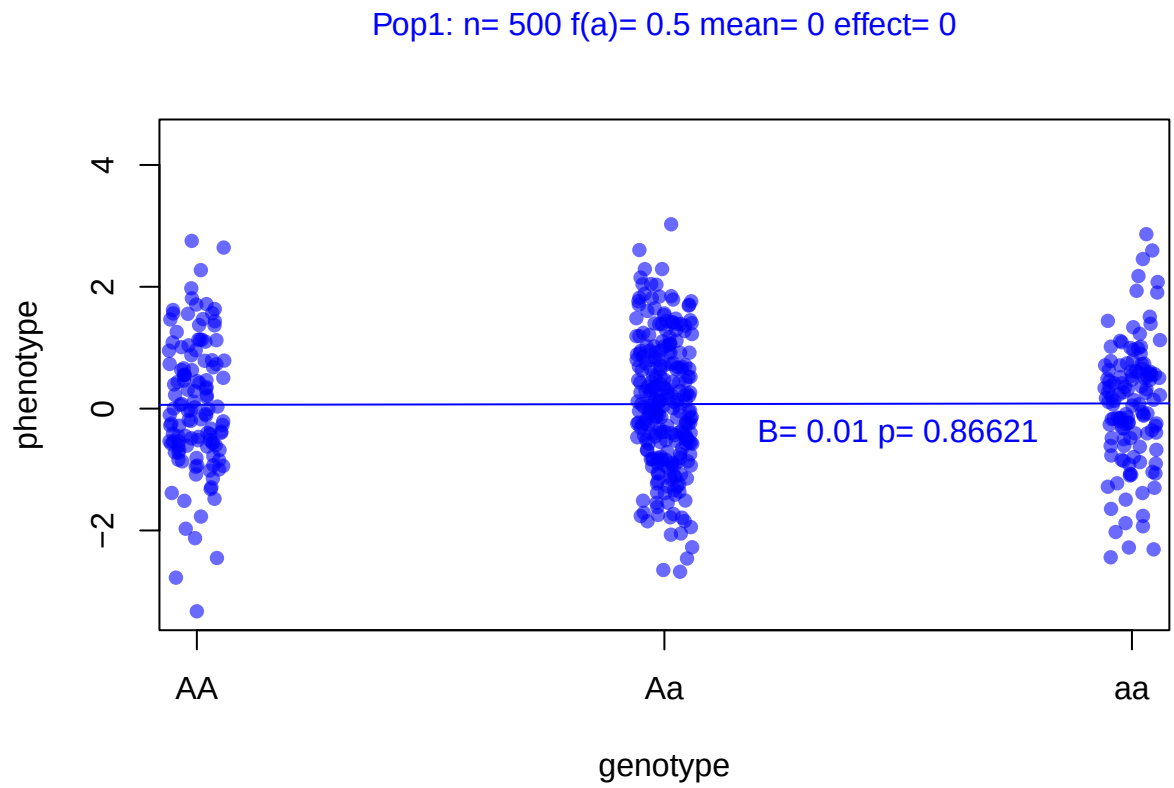
```
## Loading required package: ggplot2
```

```
#Define parameter values for two possibly distinct populations

p1=.5
mean1=0
sd1=1
effect1=0
numObs1=500
numObs2=500
p2=.9
mean2=1
effect2=0
sd2=1

# Note that as sample sizes go up, so does sensitivity to differences in effect size and allele frequen

hwprops1=as.integer(as.character(numObs1*(c(p1^2,2*p1*(1-p1),(1-p1)^2))))  #this is still screwy for lo
hwprops2=as.integer(as.character(numObs2*(c(p2^2,2*p2*(1-p2),(1-p2)^2))))

geno1<-c(rep(0,hwprops1[1]),rep(1,hwprops1[2]),rep(2,numObs1-hwprops1[1]-hwprops1[2]))
pheno1<-rnorm(numObs1,sd=1,mean=(mean1+effect1*geno1))
geno2<-c(rep(0,hwprops2[1]),rep(1,hwprops2[2]),rep(2,numObs2-hwprops2[1]-hwprops2[2]))
pheno2<-rnorm(numObs2,sd=sd2,mean=(mean2+effect2*geno2))
genoall<-c(geno1,geno2)
phenoall<-c(pheno1,pheno2)
xlim=range(genoall)
ylim=range(phenoall)
ancestry<-c(rep("Blue",numObs1),rep("Red",numObs2))


#consider the effect of the genotype in population 1

titlept1=paste("Pop1: n=",numObs1,"f(a)=",(1-p1),"mean=",mean1,"effect=",effect1)
```

```
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(0,0,255,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "blue")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

lm1<-lm(pheno1~geno1)
sum.lm1<-summary(lm1)
pval1<-print(sum.lm1$coefficients[2,4],digits=2)
```

```
## [1] 0.87
```

```
beta1<-sum.lm1$coefficients[2,1]
abline(lm1$coefficients[1],lm1$coefficients[2],col="blue")
label1<-paste("B=",round(beta1,2),"p=",round(pval1,5))
text(1.5, y = ((lm1$coefficients[1]+1.5*lm1$coefficients[2])-.5), labels = label1,col="blue")
```

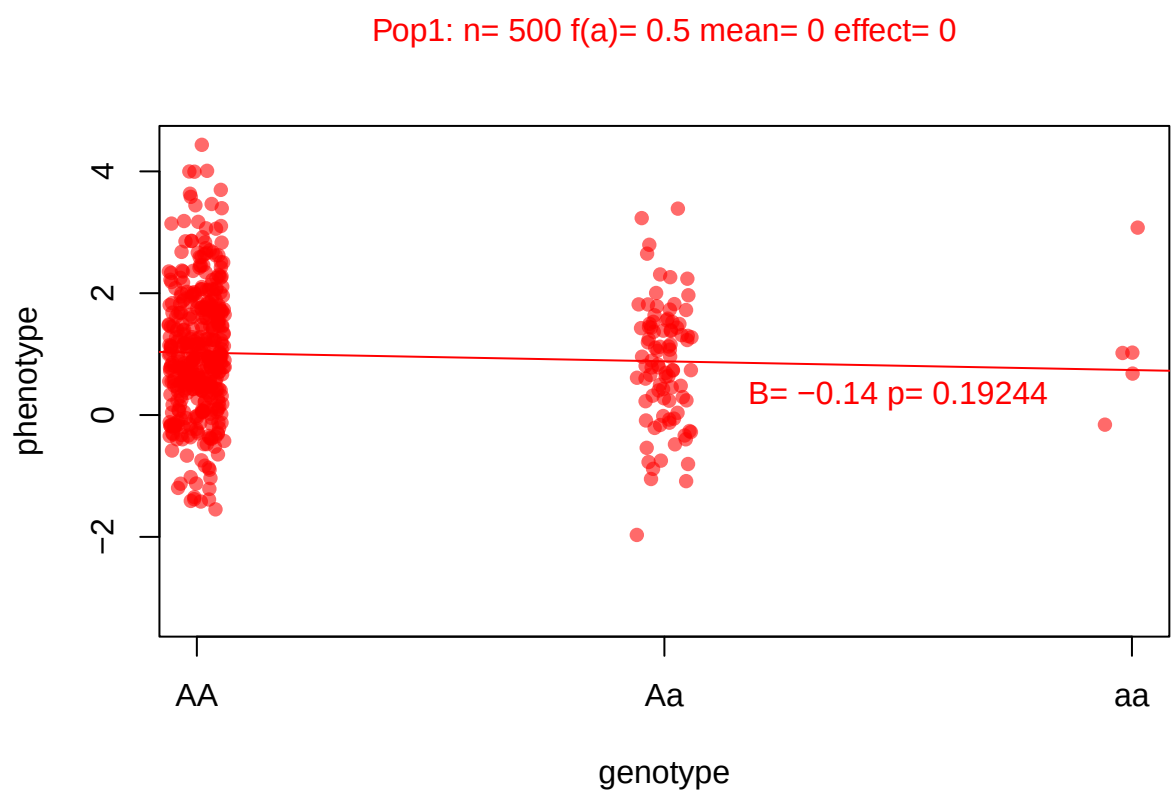



```
#consider the effect of the genotype in population 2
titlept2=paste("Pop2: n=",numObs2,"f(a)=",(1-p2),"mean=",mean2,"effect=",effect2)
plot(jitter(geno2,.3),pheno2,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(255,0,0,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "red")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

lm2<-lm(pheno2~geno2)
```

```r
sum.lm2<-summary(lm2)
pval2<-print(sum.lm2$coefficients[2,4],digits=2)
```

```
## [1] 0.19
```

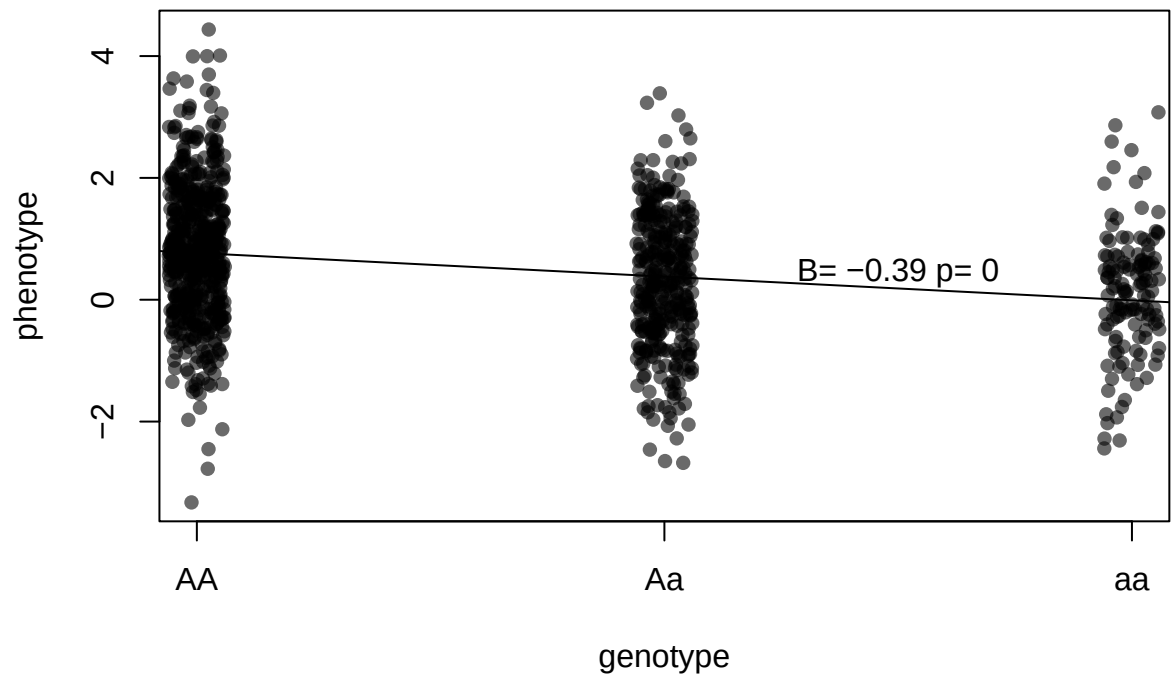
```r
beta2<-sum.lm2$coefficients[2,1]
abline(lm2$coefficients[1],lm2$coefficients[2],col="red")
label2<-paste("B=",round(beta2,2),"p=",round(pval2,5))
text(1.5, y = ((lm2$coefficients[1]+1.5*lm2$coefficients[2])-.5), labels = label2,col="red")
```



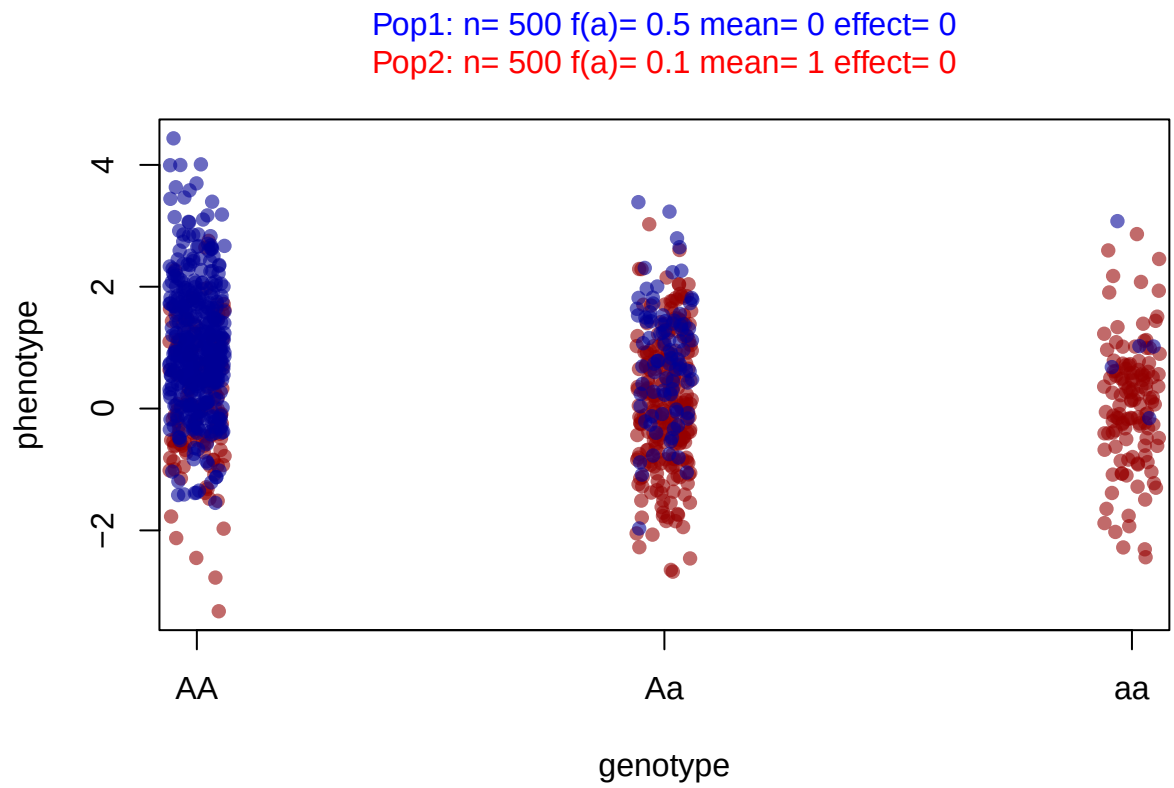

```r
#consider the effect of the genotype in population 1 and 2 jointly, ignoring any difference
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(0,0,0,150,max
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)
points(jitter(geno2,.3),pheno2,col=rgb(0,0,0,150,maxColorValue=255), pch=16)

lmall<-lm(phenoall~genoall)
sum.lmall<-summary(lmall)
pvalall<-print(sum.lmall$coefficients[2,4],digits=6)
```

```
## [1] 1.17334e-14
```

```
betaall<-sum.lmall$coefficients[2,1]


abline(lmall$coefficients[1],lmall$coefficients[2])
labelall<-paste("B=",round(betaall,2),"p=",round(pvalall,6))
text(1.5, y = ((lmall$coefficients[1]+1.5*lmall$coefficients[2])+.25), labels = labelall)
```




```
#consider the effect of the genotype in population 1 and 2 jointly, ignoring any difference but plot th

titlept1=paste("Pop1: n=",numObs1,"f(a)=",(1-p1),"mean=",mean1,"effect=",effect1)
titlept2=paste("Pop2: n=",numObs2,"f(a)=",(1-p2),"mean=",mean2,"effect=",effect2)
title12=c(titlept1,titlept2)
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(150,0,0,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "blue")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

points(jitter(geno2,.3),pheno2,col=rgb(0,0,150,150,maxColorValue=255), pch=16)
mtext(titlept2, side = 3, line = 1, cex = 1, col = "red")
```




```
#consider the effect of the genotype in population 1 and 2 jointly, accounting for difference
titlept1=paste("Pop1: n=",numObs1,"f(a)=",(1-p1),"mean=",mean1,"effect=",effect1)
titlept2=paste("Pop2: n=",numObs2,"f(a)=",(1-p2),"mean=",mean2,"effect=",effect2)
title12=c(titlept1,titlept2)
plot(jitter(geno1,.3),pheno1,xlim=xlim,ylim=ylim,xlab="genotype",ylab="phenotype",col=rgb(150,0,0,150,ma
mtext(titlept1, side = 3, line = 2, cex = 1, col = "blue")
labels=c("AA","Aa","aa")
axis(1,at=c(0,1,2),labels=labels)

points(jitter(geno2,.3),pheno2,col=rgb(0,0,150,150,maxColorValue=255), pch=16)
mtext(titlept2, side = 3, line = 1, cex = 1, col = "red")

lmall<-lm(phenoall~genoall)
sum.lmall<-summary(lmall)
pvalall<-print(sum.lmall$coefficients[2,4],digits=6)
```

```
## [1] 1.17334e-14
```

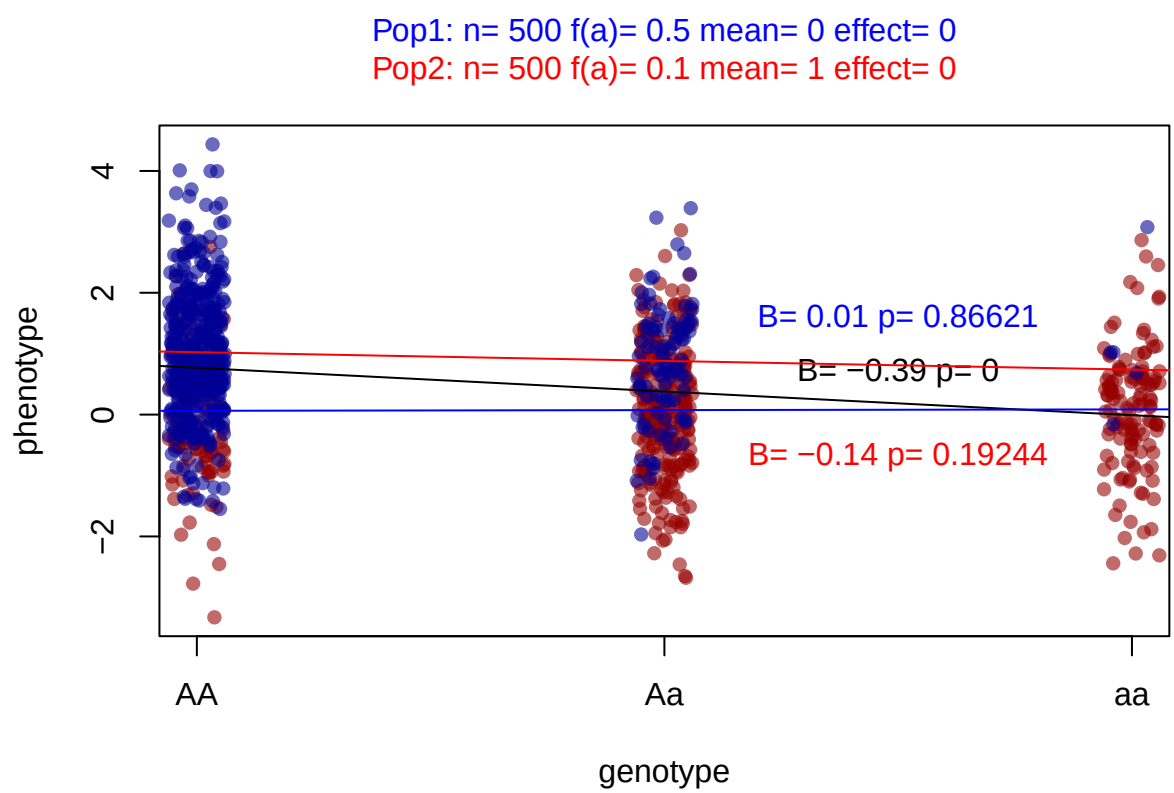
```
betaall<-sum.lmall$coefficients[2,1]

lm1<-lm(pheno1~geno1)
sum.lm1<-summary(lm1)
pval1<-print(sum.lm1$coefficients[2,4],digits=3)
```

```
## [1] 0.866
```

```r
beta1<-sum.lm1$coefficients[2,1]

lm2<-lm(pheno2~geno2)
sum.lm2<-summary(lm2)
pval2<-print(sum.lm2$coefficients[2,4],digits=3)
```

```
## [1] 0.192
```

```r
beta2<-sum.lm2$coefficients[2,1]

abline(lmall$coefficients[1],lmall$coefficients[2])
labelall<-paste("B=",round(betaall,2),"p=",round(pvalall,5))
text(1.5, y = ((lmall$coefficients[1]+1.5*lmall$coefficients[2])+.5), labels = labelall)

abline(lm1$coefficients[1],lm1$coefficients[2],col="blue")
label1<-paste("B=",round(beta1,2),"p=",round(pval1,5))
text(1.5, y = ((lm1$coefficients[1]+1.5*lm1$coefficients[2])+1.5), labels = label1,col="blue")

abline(lm2$coefficients[1],lm2$coefficients[2],col="red")
label2<-paste("B=",round(beta2,2),"p=",round(pval2,5))
text(1.5, y = ((lm2$coefficients[1]+1.5*lm2$coefficients[2])-1.5), labels = label2,col="red")
```




```r
#Ok, now that we see the issue. Lets check a model that resolves the problem
```

```r
lm1<-lm(pheno1~geno1)
summary(lm1)
```

```
##
## Call:
## lm(formula = pheno1 ~ geno1)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.3901 -0.6501 -0.0101  0.6681  2.9519
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.06281    0.08154   0.770    0.441
## geno1        0.01122    0.06658   0.169    0.866
##
## Residual standard error: 1.053 on 498 degrees of freedom
## Multiple R-squared:  5.705e-05,  Adjusted R-squared:  -0.001951
## F-statistic: 0.02841 on 1 and 498 DF,  p-value: 0.8662
```

```r
lm2<-lm(pheno2~geno2)
summary(lm2)
```

```
##
## Call:
## lm(formula = pheno2 ~ geno2)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.8497 -0.6382 -0.0747  0.6570  3.4116
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.02416    0.05134  19.949   <2e-16 ***
## geno2       -0.14286    0.10946  -1.305    0.192
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.038 on 498 degrees of freedom
## Multiple R-squared:  0.003409,   Adjusted R-squared:  0.001408
## F-statistic: 1.703 on 1 and 498 DF,  p-value: 0.1924
```

```r
lmall<-lm(phenoall~genoall)
summary(lmall)
```

```
##
## Call:
## lm(formula = phenoall ~ genoall)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -4.0954 -0.7644  0.0020  0.7362  3.6677
```

```
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.76804    0.04599  16.699  < 2e-16 ***
## genoall     -0.38872    0.04960  -7.838 1.17e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.109 on 998 degrees of freedom
## Multiple R-squared:  0.05798,    Adjusted R-squared:  0.05704
## F-statistic: 61.43 on 1 and 998 DF,  p-value: 1.173e-14
```

```r
lmallA<-lm(phenoall~genoall+ancestry)
summary(lmallA)
```

```
##
## Call:
## lm(formula = phenoall ~ genoall + ancestry)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4309 -0.6551 -0.0475  0.6739  3.4343
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.10360    0.07351   1.409    0.159
## genoall     -0.02956    0.05672  -0.521    0.602
## ancestryRed  0.89790    0.08021  11.194   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.046 on 997 degrees of freedom
## Multiple R-squared:  0.1632, Adjusted R-squared:  0.1615
## F-statistic:  97.2 on 2 and 997 DF,  p-value: < 2.2e-16
```

```r
#Summary: It's is not the genotype that is associated with the outcome, it is the ummodeled difference
```